Physics-Consistent Generative Inverse Design of Angle-Resolved Fourier-Operator Metasurfaces

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Abstract

Fourier-operator metasurfaces are usually reported task by task, for example as differentiators, edge detectors, angular filters, or polarization-selective processors. Here we focus on Angle-Resolved Fourier-Operator Metasurfaces (AFOM) and propose a general framework for joint wavelength–theta–polarization design and control. In freeform inverse design, the key object to specify and recover is the optical transfer function itself over wavelength, incidence angle, and polarization. This paper rewrites the problem in that form and develops a physics-consistent generative design loop for freeform binary metasurfaces conditioned on the target response tensor. The workflow combines fabrication-aware structure generation, rigorous simulation-based labeling on a two-channel wavelength–angle grid, surrogate-assisted candidate screening, conditional diffusion-based inverse sampling, and simulation-verified reranking under task-specific scores. Within this unified formulation, a large-NA, high-transmission, broadband second-order differentiator is used as the leading benchmark, while polarization-independent, polarization-multiplexed, fourth-order, low-pass, and spatiotemporal targets are treated as controlled variations of the same response-space inverse problem. The results show that this formulation supports high-fidelity verified designs across these diverse targets, with multiple target designs achieving the highest performance among all current designs: second-order cases reach verified task scores above 0.93 across multiple wavelengths, the low-pass benchmark attains a verified score of 0.993, and polarization-independent, polarization-multiplexed, fourth-order, low-pass, and spatiotemporal targets are all successfully realized within the same framework. The main contribution is a response-centered paradigm for metasurface inverse design in which target definition, model training, candidate generation, and final selection remain tied to the verified scattering behavior of the device.

Keywords: metasurfaces; inverse design; Fourier optics; angle-resolved response; diffusion model; full-wave simulation; polarization multiplexing

1 Introduction

The relentless pursuit of faster and more efficient computation has increasingly shifted to photonic platforms, driven by the inherent advantages of optical systems: massive parallelism, ultralow latency, and minimal energy dissipation compared to electronic counterparts [1, 2, 3]. Among these platforms, metasurfaces—subwavelength-structured nanophotonic interfaces—have emerged as transformative elements for compact optical computing because they can compress wavefront shaping, spatial-frequency filtering, polarization control, and spectral selectivity into a single patterned surface [4, 5, 6, 7, 8]. In the domain of analog image processing, metasurfaces enable direct manipulation of optical wavefronts for

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critical tasks such as edge detection, image sharpening, and feature extraction without optoelectronic conversion [9, 10, 11, 12, 13]. These capabilities form the foundation for emerging applications spanning autonomous navigation, biomedical diagnostics, and augmented/virtual reality systems [14, 15]. Compared with traditional bulky $4f$ optical systems that rely on paired lenses and spatial filters, metasurface-based processors offer significant advantages in terms of footprint, weight, and integration potential [16, 17]. Dielectric metasurfaces, in particular, can achieve high transmission efficiency while maintaining subwavelength resolution, making them ideal candidates for on-chip optical computing [18, 19].

The physical mechanism underlying metasurface-based optical computing is the engineering of the Optical Transfer Function (OTF), which describes how an optical system modulates the amplitude and phase of different spatial frequency components of the incident light field [16, 20]. In the angular spectrum representation, any input image can be decomposed into plane wave components characterized by transverse wavevectors k_x and k_y, which relate to the incidence angle θ and azimuthal angle ϕ through

$$k_x = k_0 \sin\theta \cos\phi, \qquad k_y = k_0 \sin\theta \sin\phi, \qquad k_0 = \frac{2\pi}{\lambda},$$

where k_0 is the free-space wavevector [16]. For a coherent, linear, and spatially invariant optical system, the output field $U(x, y)$ is determined by the OTF $H(k_x, k_y)$ through the Fourier transform relationship

$$U(x, y) = \mathcal{F}^{-1}\left[H(k_x, k_y) \cdot \mathcal{F}[U_0(x, y)]\right],$$

where $U_0(x, y)$ is the incident field, $\mathcal{F}$ denotes the Fourier transform, and $\mathcal{F}^{-1}$ is the inverse Fourier transform. By carefully designing the OTF $H(k_x, k_y)$, various mathematical operations can be implemented directly in the optical domain. For instance, a first-order spatial differentiator requires an OTF proportional to ik_x, which exhibits odd symmetry in the angular spectrum and enables edge detection along the corresponding direction [9, 17]. A second-order differentiator requires an OTF proportional to k_x^2, corresponding to Laplacian operations that enhance edges isotropically [10, 21]. More complex operations such as fourth-order differentiation [22] and Gaussian high-pass filtering [23] can be achieved by engineering higher-order polynomial or exponential dependencies of the OTF on the transverse wavevector.

While early metasurface designs focused primarily on achieving desired optical functions at fixed wavelengths and normal incidence, the full potential of OTF engineering lies in controlling the optical response over the complete parameter space of wavelength (λ), incidence angle (θ), and polarization (p). In this work, we introduce the concept of Angle-Resolved Fourier-Operator Metasurfaces (AFOM), where the OTF is explicitly treated as a function of these three variables:

$$\mathrm{OTF} = H(k_x, k_y; \lambda, \theta, p).$$

In this framework, various optical operations emerge as specific target geometries within this multi-dimensional response space. Spatial differentiation requires the transmission coefficient to follow specific angular dependencies:

$$t(\theta) \propto \theta, \qquad t(\theta) \propto \theta^2, \qquad t(\theta) \propto \theta^4,$$

for first-order [9], second-order [10, 12], and fourth-order differentiation [22], respectively. These operations exploit the nonlocal resonant response of carefully designed meta-atoms to create the desired polynomial OTF shape across the angular spectrum. Polarization-multiplexed operations leverage the tensor nature of the OTF, enabling independent processing for different polarization channels [24, 25]. By engineering anisotropic meta-atoms, the OTF can be designed as $H_{xx} \neq H_{yy}$, allowing orthogonal polarization components to undergo different mathematical operations simultaneously. Spatiotemporal differentiation extends the OTF concept to the frequency domain, where the metasurface response is engineered to depend on both spatial frequency (k_x) and temporal frequency (ω), enabling operations on time-varying signals [26, 13]. Convolution operators can be implemented by designing OTFs that correspond to specific kernel functions in the spatial domain, enabling more general linear image processing beyond differentiation [23]. The AFOM paradigm unifies these diverse functionalities under a common mathematical framework:

$$\mathbf{R}^{\star} = \mathbf{R}^{\star}(\lambda, \theta, p),$$

where each operation corresponds to a specific target response tensor in the wavelength–angle–polarization space, and the inverse design problem becomes finding structures that realize these target OTF geometries.

Despite the conceptual elegance of OTF engineering, the inverse design of AFOM faces fundamental challenges that limit practical realization. The mapping from metasurface structure to optical response is high-dimensional, highly nonlinear, and computationally expensive to evaluate through rigorous electromagnetic simulations [27]. For freeform binary metasurfaces with thousands of active pixels, the design space is vast, and the structure–response relationship exhibits complex dependencies. The OTF depends sensitively on geometric parameters (period, fill factor, thickness, shape) in a coupled, non-separable manner [28]. The resonant response at a given wavelength and angle is linked to the behavior across the entire spectrum through Kramers-Kronig relations and causality constraints [20]. Nonlocal metasurfaces exhibit strong angular dispersion, where small changes in incidence angle can dramatically alter the transmission phase and amplitude [11]. Anisotropic structures intended for polarization multiplexing may suffer from unwanted cross-polarization conversion that degrades operation fidelity [24]. Conventional design approaches rely on empirical trial-and-error or parametric sweeps within limited geometric families (e.g., simple gratings, hole arrays, or rod structures) [12, 13]. These methods become impractical when targeting complex OTF shapes, broadband operation, or large numerical apertures. Moreover, the inverse problem is inherently one-to-many: multiple physically distinct structures may satisfy the same target OTF within tolerance, suggesting that deterministic optimization approaches may miss superior alternative solutions [29, 30].

Machine learning methods have emerged as promising tools for metasurface inverse design, offering capabilities to learn complex structure–response mappings from data [31, 32, 33]. Existing approaches can be categorized into forward surrogate models that predict optical responses given geometric parameters [31, 34], direct inverse networks that map from target responses back to geometric parameters [32, 35], generative models including GANs [36, 37], VAEs [38], and diffusion models [39, 40] that learn the distribution of valid designs, and physics-informed approaches that incorporate electromagnetic constraints into the learning framework [41, 42]. However, these existing approaches exhibit critical limitations when applied to AFOM design. Most methods are restricted to simple, fixed structural templates with only a few tunable parameters, and cannot handle freeform binary structures with thousands of degrees of freedom where the optimal geometry may not resemble any conventional shape [31, 34]. Existing inverse design frameworks are typically trained for a single functionality and lack the flexibility to handle diverse target OTFs without retraining or architectural modification [35, 43]. The majority of prior work focuses exclusively on normal incidence or narrow angular ranges, ignoring the full angle-resolved response critical for high-NA operation and realistic imaging systems [12, 21]. Many approaches reduce the target to scalar figures of merit or sparse spectral points, discarding the structured information in the full wavelength–angle–polarization response tensor [32]. Generated designs may violate manufacturing requirements, requiring post-hoc modification that degrades performance [44].

This work addresses the limitations of existing approaches by developing a physics-consistent generative framework for freeform inverse design of AFOM. First, we construct a comprehensive dataset of fabrication-aware freeform binary metasurface structures with rigorous RCWA-simulated OTF labels spanning wavelength, incidence angle, and polarization channels. The data generation pipeline enforces physical manufacturing constraints (minimum feature size, connectivity) during structure synthesis, ensuring practical realizability of generated designs. Second, we propose a closed-loop design workflow comprising task-specific scoring functions for target OTF specification, physics-consistent data curation with adaptive quality thresholds, training of conditional diffusion models for structure generation, surrogate-assisted candidate generation with physics-guided inference, and simulation-verified reranking for final selection. This pipeline keeps the reported devices tied to verified scattering behavior instead of neural confidence alone. Third, we demonstrate that diverse optical operations—second-order differentiation, polarization-independent operation, polarization-multiplexed processing, fourth-order differentiation, low-pass filtering, and spatiotemporal differentiation—can be unified as different target geometries within the same response space. The same trained framework handles all these tasks through conditioning on the appropriate target OTF tensor $\mathbf{R}^\star(\lambda, \theta, p)$. Finally, we achieve state-of-the-art experimental benchmarks: second-order differentiators with multi-wavelength operation, high numerical aperture (NA ≈ 0.4), broadband performance, and high transmission efficiency (verified task scores above 0.93); polarization-multiplexed designs with independent differential operations for orthogonal polarizations; fourth-order and low-pass filters with complex OTF shapes (verified scores of 0.993); and spatiotemporal differentiators with joint processing of spatial and frequency information.

The remainder of this paper is organized as follows. Section 2 presents the unified response formulation and the complete physics-consistent design framework. Section 3 reports comprehensive experimental results across the benchmark tasks. Section 4 discusses scaling trends, limitations, and broader implications. Section 5 concludes.

2 Methods

2.1 Closed-loop framework

The proposed workflow is organized as a task-driven physical loop in which the generator is embedded within a broader inverse-design pipeline. A target operation is first converted into a task-aware score, and this score quantifies device performance on the desired function while defining the response object used for training, screening, and final reporting. The resulting wavelength–angle–polarization target then drives dataset construction, model training, conditional generation, and final simulation-based reranking. In symbolic form, the workflow can be written as

$$\mathbf{R}^\star \xrightarrow{\mathcal{S}} \mathcal{D}_{\text{train}} \xrightarrow{\mathcal{F},\mathcal{G}} \widehat{\mathcal{G}}_{1:K} \xrightarrow{\text{Sim.}} \widehat{\mathcal{G}}_{\text{best}}, \tag{1}$$

where $\mathbf{R}^\star$ is the target response tensor, $\mathcal{S}$ denotes the task-specific score family, which converts agreement between the target OTF and device response into a directly comparable task-performance score, $\mathcal{D}_{\text{train}}$ is the simulation-grounded training set, $\mathcal{F}$ is the forward surrogate, $\mathcal{G}$ is the conditional diffusion generator, and $\widehat{\mathcal{G}}_{1:K}$ denotes the generated candidate set. The reported output is determined by the candidate with the best verified simulation score under the chosen task, because the manuscript ultimately reports task performance rather than model-internal preference.

This choice is essential for two reasons. First, the forward and inverse models are trained against the same optical object that is later used for physical verification. Second, the candidate set remains explicit throughout inference, which is necessary because the inverse mapping from response to geometry is multimodal. The role of the forward surrogate is especially important here. The diffusion model does not directly encode electromagnetic response laws; by itself it learns structure-generation patterns from data. The forward surrogate provides an approximate structure-to-response map and injects response-level physical constraints through the surrogate-guided physics loss used during diffusion training. This loss pulls the generative process back toward regions that satisfy the target electromagnetic response, so the sampled candidates are shaped jointly by data priors and physical consistency. The pipeline therefore treats rigorous electromagnetic simulation as the physical source of truth at both training and inference time, while learned models are used to allocate solver effort, populate the feasible set efficiently, and inject response-aware constraints during training.

2.2 Dataset construction, scoring, and augmentation

The geometry class is a fabrication-oriented 64×64 freeform binary pattern with the convention $1 =$ material and $0 =$ air. The structure generator starts from a coarse random field, imposes $C_4 + \sigma_x + \sigma_y$ symmetry, and upsamples to the final resolution. This D_4-type symmetry prior is introduced to promote in-plane isotropy while retaining freeform flexibility: under this constraint, the structure or angular response over the 0°–45° sector is already sufficient to determine the full metasurface by symmetry folding. Two rounds of Gaussian smoothing followed by binarization are then used to progressively form a manufacturable freeform pattern: the first round turns the rough random field into a continuous shape prototype, while the second further regularizes boundaries and suppresses fragmented high-frequency details. To approximately enforce a minimum feature size, the code then applies morphology-based opening and closing operations. The opening step removes small protrusions, isolated islands, and overly thin bridges, whereas the closing step fills small holes and narrow cracks; if the cleanup becomes too aggressive, the procedure falls back to a more conservative closing-only variant. This produces a structure space that goes well beyond a small analytic family while remaining compatible with fabrication-oriented constraints.

All structures are labeled by RCWA under a fixed material stack consisting of a 500 nm-period silicon patterned layer of thickness 500 nm, illuminated from air and transmitting into silica. The response grid contains 11 sampled wavelengths from 800 to 1300 nm in 50 nm steps and 17 sampled angles from $-40°$ to $40°$ in $5°$ steps. Two polarization-preserving channels are retained, namely T_{pp} and T_{ss} magnitude maps, so each training condition is represented as a tensor of shape $[2, 11, 17]$. Although compact, this discretization is already sufficient to resolve wavelength drift, angular curvature, and polarization leakage across the operator families considered here.

Let $R_q(\lambda_j, \theta)$ denote the verified response slice of polarization channel $q \in \{p, s\}$ at sampled wavelength λ_j, and let

$$\widetilde{R}_q(\lambda_j, \theta) = \frac{R_q(\lambda_j, \theta)}{\max_\theta R_q(\lambda_j, \theta) + \varepsilon} \tag{2}$$

be the row-wise normalized response used for shape comparison. The role of the scoring functional family is to project each physical target onto a verified response criterion that is both task aware and numerically comparable across structures. In that sense, the score does not merely rank candidates after simulation; it acts as the explicit bridge from optical specification to learnable target.

For the second-order differentiator with large NA, high transmission, and broadband operation, the ideal angular envelope is written as

$$g_2(\theta) = \left| \frac{\sin\theta}{\sin\theta_{\max}} \right|^2, \tag{3}$$

and the row-wise score is defined by

$$S_2 = (1 - \alpha)\,(w_c S_c + w_s S_s + w_e S_e) + \alpha S_b. \tag{4}$$

Here S_c rewards suppression near $\theta = 0$, S_s measures agreement with the target k_x^2-like envelope, S_e rewards sufficiently strong outer-angle transmission, and S_b enforces persistence of the same operator behavior at neighboring wavelengths. These four terms correspond respectively to low-spatial-frequency suppression, correct differentiator-like angular growth, preservation of high-spatial-frequency content, and finite-bandwidth robustness. Other task-specific scores are built on the same response-centered principle and are detailed in the Supporting Information.

2.3 Forward surrogate

The forward surrogate learns the mapping

$$\mathcal{F}_\phi : \mathbf{G} \in \{0,1\}^{64\times 64} \mapsto \widehat{\mathbf{R}} \in \mathbb{R}^{2\times 11\times 17}, \tag{5}$$

where $\widehat{\mathbf{R}}(:, j, k)$ denotes the joint two-polarization prediction at the jth wavelength sample and the kth angle sample. Its role is intentionally limited but physically important: it provides a fast approximation to the structure-to-response map for large-scale candidate screening, supplies the response-consistency term used to guide diffusion training, and reallocates RCWA effort toward candidates that are more likely to satisfy the target response.

Architecturally, the model is a coordinate-aware convolutional encoder followed by multiscale response-grid fusion. Two Cartesian coordinate channels are concatenated to the binary geometry so that the network can relate planar layout to angular scattering directionality rather than treating the pattern as a translation-invariant texture. Residual convolutional blocks then encode the structure across progressively coarser scales, which is important because local subwavelength boundaries and global topology influence the response in different ways. Instead of decoding back to the image plane, all feature levels are projected directly onto the 11×17 wavelength–angle grid and fused there, so the prediction head operates on the same physical sampling manifold used for RCWA labels.

The deepest latent representation is further compressed by a global multilayer perceptron and broadcast to the whole response grid, injecting device-level information such as overall topology and effective fill into every wavelength–angle location. The response grid itself is also endowed with explicit coordinates, allowing the network to distinguish central from outer angles and short from long wavelengths. Finally, the two polarization-preserving channels are predicted jointly rather than separately, because T_{pp} and T_{ss} originate from the same geometry and their correlation or leakage is itself part of the physical specification.

For training, the response tensor is normalized pointwise across channel, wavelength, and angle, invalid simulation samples are removed before statistics are computed, and augmentation is restricted to symmetry-preserving flips that leave the optical response unchanged. The surrogate is optimized with an L_1 loss in normalized response space,

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$$\mathcal{L}_{\text{fwd}} = \left\| \mathcal{F}_\phi(\mathbf{G}) - \widetilde{\mathbf{R}} \right\|_1, \tag{6}$$

while physical-space MAE is tracked during validation after denormalization. In the retained training log, the normalized validation loss decreases from about 0.63 at initialization to about 0.366 near the end of training, which is sufficient for the model to act as a practical screening oracle even though final claims are still deferred to full-wave simulation.

2.4 Conditional diffusion inverse design

The inverse model is a conditional diffusion process that takes the target response tensor as condition and samples binary structure candidates. During training, binary layouts are remapped from $[0, 1]$ to $[-1, 1]$, a cosine noise schedule is used, and the model is trained in the velocity parameterization v. Let x_0 denote the clean structure, x_t the noisy sample at time step t, and $\mathbf{c}$ the normalized target response tensor. The forward noising process is

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I), \tag{7}$$

and the network learns $v_\theta(x_t, t, \mathbf{c})$ under a diffusion loss

$$\mathcal{L}_{\text{diff}} = \left\| v_\theta(x_t, t, \mathbf{c}) - v^\star(x_0, \epsilon, t) \right\|_2^2. \tag{8}$$

The backbone is a conditional U-Net with three design choices that are particularly important for this problem. First, the condition is itself a structured two-channel response field, so the model uses a dedicated condition encoder in place of a simple class-label embedding. This encoder maps the target tensor into a global condition embedding, a hierarchy of condition feature maps, and a set of spectral tokens, so the same response specification is available in both compact and spatially structured forms. Second, the global embedding is injected into residual blocks together with the time embedding, while the condition feature maps are fused into the corresponding decoder and encoder resolutions; in parallel, cross-attention injects the spectral tokens into the latent spatial hierarchy, allowing local geometry evolution to remain coupled to the global wavelength–angle response pattern throughout denoising. Third, classifier-free guidance is enabled by condition dropout during training and by guided sampling during inference.

Pure diffusion loss is not enough for this application, because a visually plausible binary layout can still be physically poor. The training objective therefore adds two auxiliary terms:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{diff}} + \lambda_{\text{phys}}\mathcal{L}_{\text{phys}} + \lambda_{\text{bin}}\mathcal{L}_{\text{bin}}, \tag{9}$$

where $\mathcal{L}_{\text{phys}} = \|\mathcal{F}_\phi(\widehat{x}_0) - \mathbf{c}\|_1$ encourages the denoised prediction to preserve the target response under the forward surrogate, and $\mathcal{L}_{\text{bin}}$ penalizes uncertain gray values through $x(1-x)$. In this formulation, $\mathcal{L}_{\text{phys}}$ is the mechanism that injects response-level physical constraints into diffusion training through the surrogate-predicted electromagnetic behavior. In the retained training setup, $\lambda_{\text{phys}} = 0.5$ and $\lambda_{\text{bin}} = 0.05$. The final saved validation loss is around 0.129, with the physics term remaining a nontrivial contributor throughout training. This supports the interpretation that the inverse model is being pushed toward physically meaningful candidates instead of merely toward dataset likeness.

2.5 Physics-guided inference and simulation-based reranking

Inference is deliberately candidate-based. For each target case, an ideal target OTF over (λ, θ, p) is constructed, normalized with the saved statistics, and repeated K times to condition batch sampling. The diffusion model then produces multiple binary candidates with classifier-free guidance. The surrogate supplies an inexpensive first-pass ranking, after which selected candidates are pushed back through full-wave simulation under the exact wavelength and angle points relevant to the task.

The final metric is task specific but always evaluated in verified response space. For example, the second-order score combines center suppression, agreement with the ideal even-symmetric angular profile, and outer-angle support. The polarization-independent score adds channel matching. The polarization-multiplexed score rewards the active channel while suppressing the passive one. The spatiotemporal score evaluates the full wavelength–angle window instead of a single row. This division of labor is the practical definition of physics consistency in the paper: neural models generate and pre-rank candidates, while verified scattering responses determine what is ultimately reported as the design outcome.

3 Results

3.1 Second-Order Differentiators with Large NA, High Transmission, and Broadband Operation

This second-order benchmark is emphasized here because it simultaneously requires large numerical aperture, high transmission, broadband operation, response curvature control, center suppression, outer-angle support, and practical image-processing behavior. The verified second-order score is evaluated wavelength by wavelength on the T_{pp} response row. The resulting score landscape is highly nonuniform, with mean scores around 0.15–0.17 over the full dataset but top candidates rising above 0.93 at 1050 and 1100 nm. This gap is important: it shows that the feasible set is sparse but not empty, exactly the regime in which generative candidate exploration is valuable.

At 1050 and 1100 nm, the top verified second-order candidates both reach scores slightly above 0.93, and a second high-quality family is also visible near 1150 nm with scores above 0.925. These leading devices were not selected for one isolated metric. They are highlighted because they jointly exhibit the operator-like bowl around $\theta = 0$, strong growth toward large $|\theta|$, and robust amplitude at high angles.

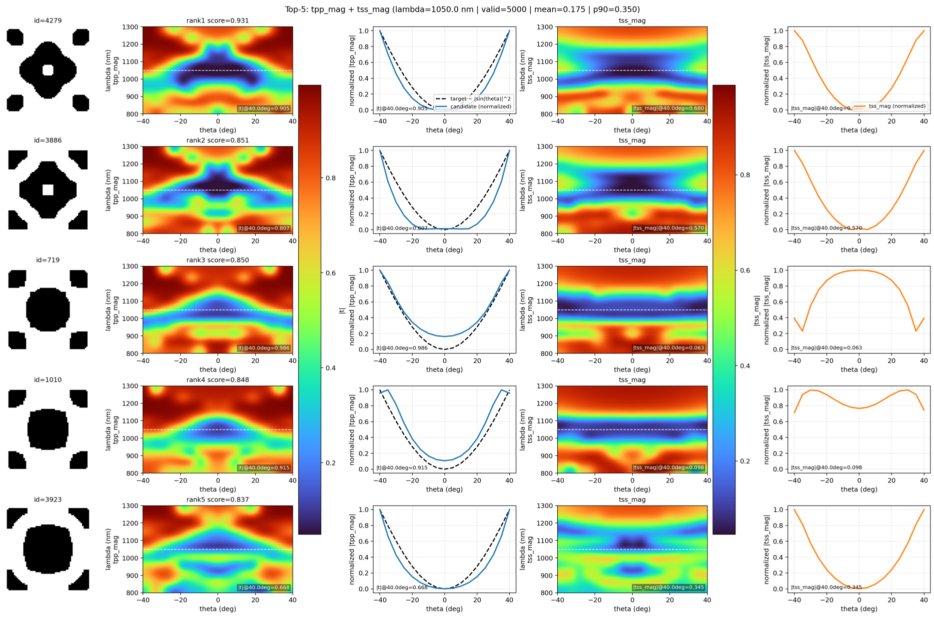

Figure 1: Top ranked second-order candidates at 1050 nm. The ranking overview visualizes the concentration of operator-like responses that motivates task-aware simulation-based reranking.

Figure 1 shows the top-five ranking overview at 1050 nm, where the leading candidate already displays a representative second-order response. The result is more than a "high score by normalization." It preserves the expected second-order envelope over a wide angular aperture and therefore remains meaningful under operator-level interpretation. The recurrence of closely related high-performing responses across independent evaluations suggests that the best second-order devices occupy especially stable regions of the verified response manifold.

The second-order case also supports the strongest literature-level positioning. Under the same downstream imaging analysis used throughout this study, the best 1050 nm result is associated with a normalized task score of about 0.988, a relative edge amplitude ratio near 0.97, an effective working bandwidth around 100 nm, and a numerical aperture around 0.64. These numbers place the design in the same performance conversation as recent broadband, high-NA, high-efficiency differentiator reports [25, 24, 22, 21]. The precise contribution of the present work, however, is different from a single-device record chase: the same response-centered framework produces this high-performing second-order behavior and also extends naturally to polarization and spatiotemporal targets.

Figure 2 closes the argument by moving from response space to image space. Once the verified transfer function is inserted back into the Fourier imaging pipeline, the resulting output exhibits the expected second-derivative effect: low-frequency regions are suppressed while edge-like features and curvature-sensitive structures are enhanced. This is the key bridge between operator scoring and practical device function. The inverse-design loop is successful because a physically verified structure reproduces both the target response geometry and its intended optical action.

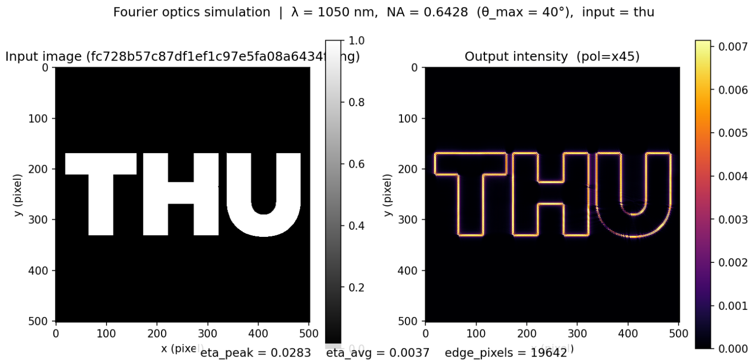

Figure 2: Image-domain validation for the leading 1050 nm second-order result. The response-level agreement translates into practical edge-enhancement behavior rather than remaining a numerical score gain only.

Table 1: Representative verified results used throughout the paper. Scores reflect task-specific rankings rather than abstract model confidence.

Task	λ (nm)	Main score	Notes
Second-order	1050	0.9306	representative T_{pp} response with large NA and high transmission
Second-order	1100	0.9308	strongest verified 1100 nm result
Polarization independent	1000	0.9358	matched dual-channel operator response
Polarization multiplexed	1100	0.7283	active score 0.9348 with passive suppression retained
Fourth-order	1150	0.8884	sharp outer-angle growth
Low-pass	1050	0.9934	strongest passband-shaping result
Spatiotemporal window	950	0.7271	strongest merged window score in the verified screen

3.2 Generalization to other OTF targets

The most important question after this second-order result is whether the same response-centered loop continues to behave coherently when the target shape changes. The verified results give a positive answer across three independent axes.

The first axis is polarization. In the polarization-independent benchmark, the best 1000 nm result reaches a joint verified score of 0.9358, while the leading 1050 and 1100 nm results remain above 0.88 and 0.91, respectively. This matters because the task score does not merely average two channels: it rewards channel-wise operator fidelity and explicit amplitude agreement around $\pm 40°$. The fact that the same framework can produce matched T_{pp} and T_{ss} operator responses shows that polarization enters naturally as a first-class conditioning axis.

The second axis is polarization multiplexing. The strongest verified multiplexed result at 1100 nm reaches a joint score of 0.7283, together with an active-channel operator score of 0.9348 and a passive-channel suppression term of 0.5218. In other words, the framework learns to preserve operator shape while also distinguishing which channel should carry the response and which should remain as silent as possible. This is exactly the kind of channel-level tradeoff that a unified response tensor should expose.

The third axis is target-shape diversity. In the fourth-order benchmark, the 1050 nm response family reaches a top score of 0.9222, while the strongest verified 1150 nm result reaches 0.8884. These values

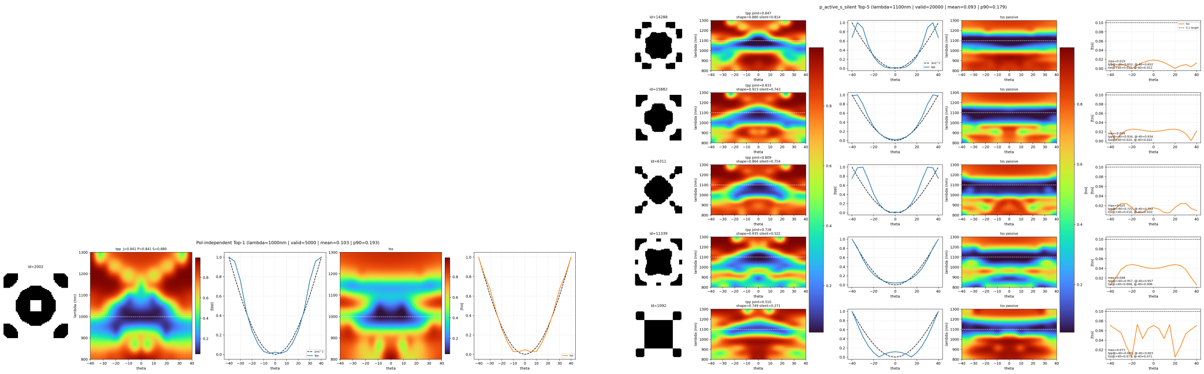

Figure 3: Polarization-aware generalization. Left: polarization-independent top case at 1000 nm. Right: polarization-multiplexed top candidates at 1100 nm. The same inverse-design loop can either align the two channels or intentionally separate them.

indicate that the framework remains effective even when the desired response becomes sharper than a second-order bowl. In the low-pass benchmark, the best verified results reach 0.9934 at 1050 nm and 0.9863 at 1100 nm, demonstrating that the same tensor-conditioning strategy can realize a fundamentally different angular envelope in addition to polynomial-like operator responses.

Spatiotemporal targets provide the strongest stress test because they are evaluated on a two-dimensional wavelength–angle window instead of a single wavelength row. The dataset-level screen at 950 nm identifies a top merged score of 0.7271, which is already meaningful given the difficulty of jointly controlling the central zero lines and the corner lobes of the ideal map. This confirms that the response-window view is not merely rhetorical, but remains quantitatively meaningful under the same verified ranking logic.

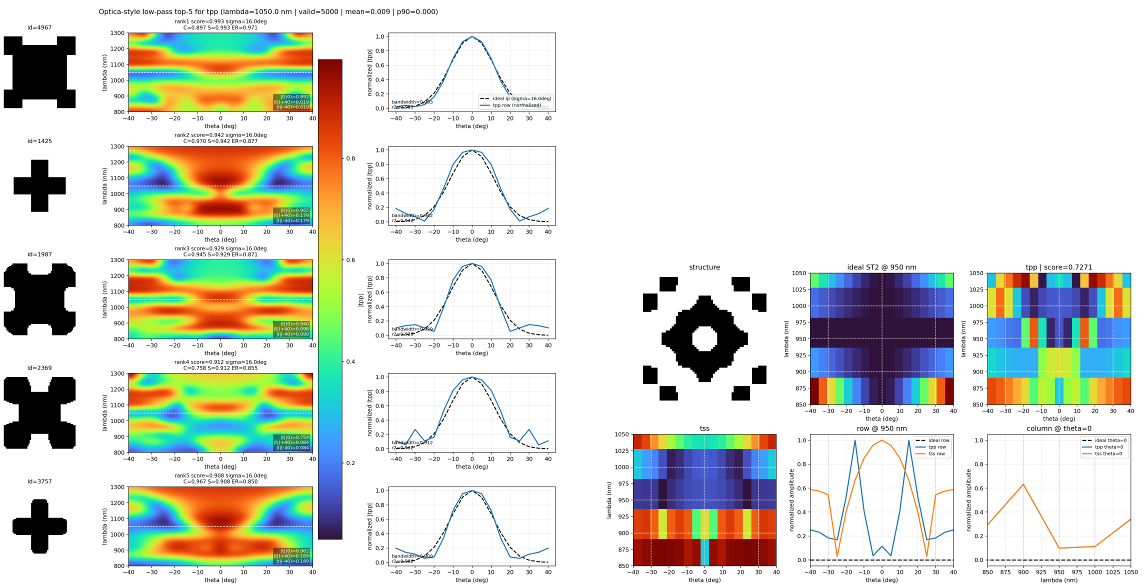

Figure 4: Generalization beyond second-order differentiation. Left: low-pass target at 1050 nm. Right: top spatiotemporal candidate at 950 nm from the wavelength–angle window ranking pipeline.

Taken together, Figures 3 and 4 support the same conclusion. The present method goes beyond a differentiator-specific recipe. It forms a unified response-space design loop that can absorb changes in channel requirements, target curvature, target envelope, and even in the dimensionality of the target support.

3.3 Model comparison and evaluation protocol

A unified comparison protocol is defined for topology optimization, CVAE, cGAN, and diffusion under the same target condition and the same simulation-based post-verification metrics. The shared evaluation reports best score, mean score, top-3 score, success rate above threshold, spectrum MAE, binary rate, diversity, and inference time. This is the correct comparison language for the present problem because a strong inverse designer should hit good samples consistently and should also populate the feasible set with diverse physically useful candidates.

At the current stage of the study, the retained checkpoints directly support the forward surrogate and the physics-guided diffusion pipeline, whereas the cVAE and cGAN baselines and the topology-optimization branch are not all accompanied by synchronized checkpoints. Table 2 therefore serves two purposes. It makes the quantitative protocol explicit and it records only the directly available evidence without fabricating missing baseline numbers.

Table 2: Comparison protocol used in this study. The metrics are computed after full-wave verification. Quantitative training endpoints are listed only where directly retained evidence is available.

Method	Condition / objective	Direct retained evidence	Status
Forward surrogate	structure → response, normalized L_1	val loss ≈ 0.366	checkpoint + log retained
Diffusion	response → structure, diffusion + physics + binary losses	val loss ≈ 0.129	checkpoint + log retained
CVAE	response → structure, variational baseline	unified eval supported	checkpoint absent locally
cGAN	response → structure, adversarial baseline	unified eval supported	checkpoint absent locally
Topo-opt	task-guided iterative search	unified eval supported	legacy dependency incomplete locally

Even with this limitation, the comparison section remains scientifically useful. The evaluation protocol already formalizes the right question: the inverse method should be judged by verified best-case score, average candidate quality, response MAE, binarity, and diversity under shared targets. The present physics-guided diffusion pipeline is the only branch for which the end-to-end checkpointed evidence is directly retained, which is why it becomes the center of the manuscript.

4 Discussion

The first point worth emphasizing is a scaling-law trend. In the present study, performance improves as the verified dataset grows from the 1 000-sample level to the 2×10^4 level, and it also benefits from stronger surrogate and generative backbones. In that sense, the present results already provide an initial scaling-law signal of their own. The analogy to empirical scaling laws in large language models is suggestive rather than literal [45]: once the target is a structured wavelength–angle–polarization field, success depends on how well the model sees rare but valuable response modes and how accurately the forward surrogate resolves them. The main limitation is that the current scaling axis is still narrow, since the framework so far fixes the material platform, lattice period, and amplitude-only response representation. A natural next step is therefore to scale not only dataset size and model capacity, but also the physical design space itself, by incorporating multiple material systems, variable periods, and phase-sensitive targets into the same response-centered formulation. If those axes can be enlarged together with denser electromagnetic verification, the framework may evolve toward a more general metasurface inverse-design regime.

The second point is the extensibility of the design space itself. Because the target is written as a response tensor rather than a single scalar objective, the same framework can in principle be extended toward joint polarization–frequency–angle design, where multiple physical channels are shaped simultaneously. This broader setting is relevant to compact folded and angle-resolved spectrometers [46, 47], multidimensional sensing and computational photodetection [48], and polarization-encoded information or optical encryption devices [49]. In that sense, the present work is better viewed as a general response-level design methodology than as a closed list of six isolated examples.

5 Conclusion

This paper has rewritten freeform Fourier-operator metasurface design as a response-centered inverse problem defined over wavelength, angle, and polarization. On that basis, it develops a coherent physical loop comprising fabrication-aware structure generation, simulation-based labeling, score-aware data preparation, forward surrogate training, conditional diffusion sampling, and simulation-based reranking. The key scientific point is that the optical target should be specified in the same response space in which the final device is verified, placing the present study in closer contact with objective-driven photonic inverse design and response-faithful electromagnetic modeling [35, 38, 44].

Within this formulation, a second-order differentiator with large NA, high transmission, and broadband operation reaches a leading level of verified performance across multiple wavelengths, while polarization-independent, polarization-multiplexed, fourth-order, low-pass, and spatiotemporal targets can likewise be handled at a competitive level within the same unified inference framework. The broader contribution is the demonstration that multiple operator families can be described, generated, and judged within one physically consistent design language.

The broader implication is methodological. Freeform metasurface inverse design is better treated as target-conditioned feasible-set discovery under electromagnetic verification than as single-geometry regression or task-label prediction detached from the transfer function. The present study provides one concrete realization of that principle, and it points naturally toward future extensions involving richer polarization control, full complex-response targets, broader physical design spaces, tighter fabrication models, and larger-scale physics-guided datasets.

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